

Day – 4 Introduction to OSPF Theory

Task – 1 Create a simple network in cisco.

- Step 1: Place the Routers
- Open Cisco Packet Tracer. Click on Routers in the bottom-left device category box.
- Select three routers and place them on the workspace in a triangle formation (naming them R1, R2, and R3 for clarity).
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- Step 2: Connect the Topology
- Click on the Connections icon (the lightning bolt symbol) in the bottom-left menu. Power off the router, add a serial interface module (like HWIC-2T), power it back on, and use the Serial DCE cable to connect them.
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- Step 3: Assign IP Addresses & Routing
- Click on each router, go to the CLI tab, and assign IP addresses to the connected interfaces (e.g., subnet 192.168.12.0/30 between R1 and R2).
- Enable the interfaces using the no shutdown command.
- Configure routing (either Static Routes or a dynamic routing protocol like OSPF) so all routers can reach one another across the loop.
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Task – 2 Assign ip to all the devices and checking ping.

- IP Addressing Scheme (/30 Subnets)
- Link R1 to R2: 192.168.12.0/30 (Subnet mask: 255.255.255.252)
- R1 (GigabitEthernet0/0): 192.168.12.1
- R2 (GigabitEthernet0/0): 192.168.12.2
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- Link R2 to R3: 192.168.23.0/30 (Subnet mask: 255.255.255.252)
- R2 (GigabitEthernet0/1): 192.168.23.1
- R3 (GigabitEthernet0/0): 192.168.23.2
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- Link R3 to R1: 192.168.31.0/30 (Subnet mask: 255.255.255.252)
- R3 (GigabitEthernet0/1): 192.168.31.2
- R1 (GigabitEthernet0/1): 192.168.31.1

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- Step 1: Configure Router 1 (R1)
- Open R1, go to the CLI tab, and enter the following commands:
- enable
- configure terminal
- hostname R1
- interface GigabitEthernet0/0
- ip address 192.168.12.1 255.255.255.252
- no shutdown
- exit
- interface GigabitEthernet0/1
- ip address 192.168.31.1 255.255.255.252
- no shutdown
- Exit
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- Step 2: Configure Router 2 (R2)
- Open R2, go to the CLI tab, and enter the following commands:
- enable
- configure terminal
- hostname R2
- interface GigabitEthernet0/0
- ip address 192.168.12.2 255.255.255.252
- no shutdown
- exit
- interface GigabitEthernet0/1
- ip address 192.168.23.1 255.255.255.252
- no shutdown
- exit
- Step 3: Configure Router 3 (R3)
- Open R3, go to the CLI tab, and enter the following commands:
- enable
- configure terminal
- hostname R3
- interface GigabitEthernet0/0
- ip address 192.168.23.2 255.255.255.252
- no shutdown
- exit
- interface GigabitEthernet0/1
- ip address 192.168.31.2 255.255.255.252
- no shutdown
- exit

- Step 4: Verify Connectivity with Ping
- Once the link indicator lights turn green, you can test direct connectivity between routers using the privileged EXEC mode (>).
- From R1, ping its direct
 - ping 192.168.12.2 (Pinging R2)
 - ping 192.168.31.2 (Pinging R3)
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- From R2, test :
 - ping 192.168.12.1 (Pinging R1)
 - ping 192.168.23.2 (Pinging R3)
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Task – 3 Config the OSPF on each router.

- Step 1: Configure OSPF on Router 1 (R1)
- Open R1, go to the CLI, and run the following:
 - configure terminal
 - router ospf 1
 - network 192.168.12.0 0.0.0.3 area 0
 - network 192.168.31.0 0.0.0.3 area 0
 - End
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- Step 2: Configure OSPF on Router 2 (R2)
- Open R2, go to the CLI, and run the following:
 - configure terminal
 - router ospf 1
 - network 192.168.12.0 0.0.0.3 area 0
 - network 192.168.23.0 0.0.0.3 area 0
 - End
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- Step 3: Configure OSPF on Router 3 (R3)
- Open R3, go to the CLI, and run the following:
 - configure terminal
 - router ospf 1
 - network 192.168.23.0 0.0.0.3 area 0
 - network 192.168.31.0 0.0.0.3 area 0
 - End
- At last checking the ping.

Task – 4 Checking the output of second router of OSPF.

- When using Ethernet (GigabitEthernet) interfaces in Cisco Packet Tracer, OSPF operates in a Broadcast network type by default. This means the routers will automatically elect a Designated Router (DR) and a Backup Designated Router (BDR) for each segment to manage routing updates efficiently.

- 1. Router 1 (R1) Adjacency Observation
- Open R1, enter privileged EXEC mode (enable), and run the command:
- R1# show ip ospf neighbor
- | Neighbor ID | Pri | State | Dead Time | Address | Interface |
|--------------|-----|----------|-----------|--------------|--------------------|
| 192.168.12.2 | 1 | FULL/BDR | 00:00:34 | 192.168.12.2 | GigabitEthernet0/0 |
| 192.168.31.2 | 1 | FULL/DR | 00:00:38 | 192.168.31.2 | GigabitEthernet0/1 |
- Observation: R1 has successfully formed a FULL adjacency with both R2 and R3. Notice the DR/BDR roles assigned on its respective interfaces.
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- 2. Router 2 (R2) Adjacency Observation
- Open R2, enter privileged EXEC mode, and run the command:
- R2# show ip ospf neighbor
- | Neighbor ID | Pri | State | Dead Time | Address | Interface |
|--------------|-----|----------|-----------|--------------|--------------------|
| 192.168.12.1 | 1 | FULL/DR | 00:00:32 | 192.168.12.1 | GigabitEthernet0/0 |
| 192.168.23.2 | 1 | FULL/BDR | 00:00:35 | 192.168.23.2 | GigabitEthernet0/1 |
- Observation: R2 also shows FULL states with R1 and R3, confirming that bidirectional communication and database synchronization are complete.
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- 3. Router 3 (R3) Adjacency Observation
- Open R3, enter privileged EXEC mode, and run the command:
- R3# show ip ospf neighbor
- | Neighbor ID | Pri | State | Dead Time | Address | Interface |
|--------------|-----|----------|-----------|--------------|--------------------|
| 192.168.23.1 | 1 | FULL/DR | 00:00:31 | 192.168.23.1 | GigabitEthernet0/0 |
| 192.168.31.1 | 1 | FULL/BDR | 00:00:33 | 192.168.31.1 | GigabitEthernet0/1 |
- Observation: R3 completes the triangle loop with active FULL states on both of its interfaces.

Task – 5 Changing the OSPF cost by using command.

- Changing the OSPF cost allows you to manipulate path selection, as OSPF prefers routes with the lowest total cumulative cost to reach a destination network.
- Step 1: Change the OSPF Cost on R1

- Open R1, go to configuration mode, and increase the cost on the interface connecting to R2 (GigabitEthernet0/0) from the default to 50:
- configure terminal
- interface GigabitEthernet0/0
- ip ospf cost 50
- End
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- Step 2: Verify the Routing Table Changes
- To see how this adjustment affects the paths chosen by OSPF, open R1 and run the show ip route ospf command.
- R1# show ip route ospf
- 192.168.23.0/30 is subnetted, 1 subnets
- 192.168.23.0 [110/51] via 192.168.31.2, 00:02:15, GigabitEthernet0/1
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